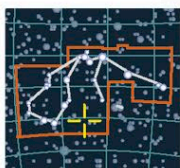


PLANETARY NEBULA

Helix Nebula



CATALOG NUMBER
NGC 7293

DISTANCE FROM SUN
Up to 650 light-years

MAGNITUDE 6.5

AQUARIUS

The Helix Nebula is the closest planetary nebula to the Sun, but its actual distance is uncertain, and estimates vary from 85 to 650 light-years. It is called the Helix Nebula because, from Earth, the outer gases of the star expelled into space give the impression that we are looking down the length of a helix. One of the largest known planetary nebulae, its main rings are about 1.5 light-years in diameter and span an apparent distance of more than half the width of a full moon. Its outer halo extends up to twice this distance. The dying star at the center of the nebula is destined to become a white dwarf, and as it continues to use up all its energy, it will continue to expel material into the interstellar medium. The Helix Nebula presents an impressive example of the final stage that stars like our Sun will experience before collapsing for the last time. It was first discovered by the German astronomer Karl Ludwig Harding in around 1824, and its size and proximity mean that it has been extensively observed and imaged.

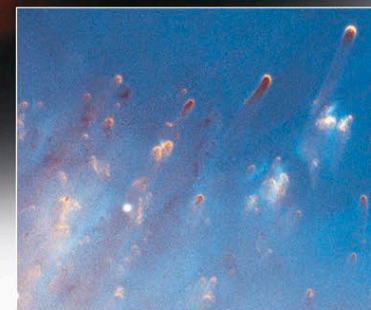
GLOWING HALO

Rings of expelled material glow red in the light produced by nitrogen and hydrogen atoms when they are energized by ultraviolet radiation.

Detailed images made of the inner edge of the ring of material surrounding the central star have shown "droplets" of cooler gas twice the diameter of our solar system, radiating outward for billions of miles. These were probably formed when a fast-moving shell of gas, expelled by the dying star, collided with slower-moving material thrown off thousands of years before.

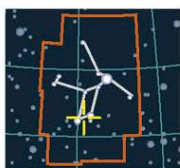
COMETLIKE KNOTS

Resembling comets, these tadpole-shaped gaseous knots are several billion miles across. They lie like spokes in a wheel along the inner edge of the ring of ejected gas surrounding the central star.



PLANETARY NEBULA

Ring Nebula



CATALOG NUMBER
M57

DISTANCE FROM SUN
2,000 light-years

MAGNITUDE 8.8

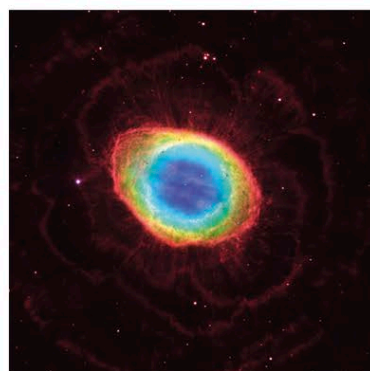
LYRA

One of the best known planetary nebulae, the Ring Nebula was discovered in 1779 by French astronomer Antoine Darquier de Pellepoix. When seen through a small telescope, it appears larger than the planet Jupiter. Its central star, a planet-sized white dwarf of only about 15th magnitude, was not discovered until 1800, when it was found by German astronomer Friedrich von Hahn. There has been a great deal of discussion about the true shape of the Ring Nebula. Although it appears like a flattened ring, some astronomers believe the stellar material has been expelled in a spherical shell that only looks like a ring because we view it through a thicker layer at its edges. Others believe it is a torus (shaped like a ring doughnut), which would look similar to the Dumbbell Nebula if viewed side-on, or that it is cylindrical or tubelike. The nebula

appears to be about 1 light-year in diameter, but it has an outer halo of material that extends for more than 2 light-years. This is possibly a remnant of the central star's stellar winds before the nebula itself was ejected. The nebula is lit by fluorescence caused by the large amount of ultraviolet radiation emitted by the central star. The rate of the ring's expansion indicates that the nebula started to form about 20,000 years before it was as it appears to us now.

COMPLEX AND COLORFUL

This unique composite image brings out details of the Ring Nebula, including a surrounding many-lobed hydrogen cloud and the blue glow of helium energized by radiation from the white dwarf at its center.



PLANETARY NEBULA

Twin Jet Nebula



CATALOG NUMBER
M2-9

DISTANCE FROM SUN
2,100 light-years

MAGNITUDE 14.7

OPHIUCHUS

The Twin Jet Nebula is one of the most striking examples of a butterfly or bipolar planetary nebula. It is believed that the star at the center of this nebula is actually a binary pair with a period of about 100 years, whose complex interplay has affected the shape of the resulting planetary nebula. The gravitational interaction between the stars has pulled stellar

material around them into a dense disk with a diameter about 10 times that of Pluto's orbit. About 1,200 years before this happened, one of the stars had an outburst, ejecting material in a strong stellar wind. This rammed into the disk, which acted like a nozzle, deflecting the material in perpendicular directions, forming the two lobes stretching out into space. This is very similar to the process that takes place in jet propulsion engines. Studies have suggested that the nebula's size has increased steadily with time and that the material is flowing outward at up to 450,000 mph (720,000 kph).

BUTTERFLY WINGS

This false-color image reveals some of the elements present in the nebula's lobes—red indicates sulfur, while blue highlights oxygen.

